Mtkvari Hydropower Project in Georgia – Dam safety upgrade of a project under execution

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The Mtkvari Hydroelectric Power Plant (HPP) is under construction on the Mtkvari river in Georgia. This scheme is of strategic importance for the development of a stable, reliable, cost-effective and efficient national electricity supply. This paper provides the main features of this project, as well as at the technical challenges and their mitigation to achieve full compliance with good international practice.

While over the last decade the hydropower sector has contributed to almost 79% of Georgia's electricity generation, the government's energy strategy is to double its hydropower capacity over the next decade to secure energy supply, develop cross-border electricity trade opportunities, and reduce reliance on fossil fuel imports by tapping the still significant unexploited hydropower potential of the country.
One of the major hydropower developments in Georgia is the 53 MW Mtkvari HPP by Mtkvari Hesi LLC, which is expected to produce 260 GWh per year in average. The project is located on the Mtkvari river in the Aspindza and Akhaltsikhe districts of the Samtskhe-Javakheti region in Southern Georgia, close to the Turkish border.
Construction activities are ongoing and are planned to be completed by the end of 2024, with the commissioning during the first quarter of 2025.

1 Overview

1.1 Project background

The project was launched in 2009. In 2021, Gruner Stucky was appointed as the Owner's Engineer, responsible for construction supervision activities, design review and technical assistance. One of the main tasks was the review of the initial design concept, including the structures already built, and to define optimisation measures to allow for the design to comply with good international practice.

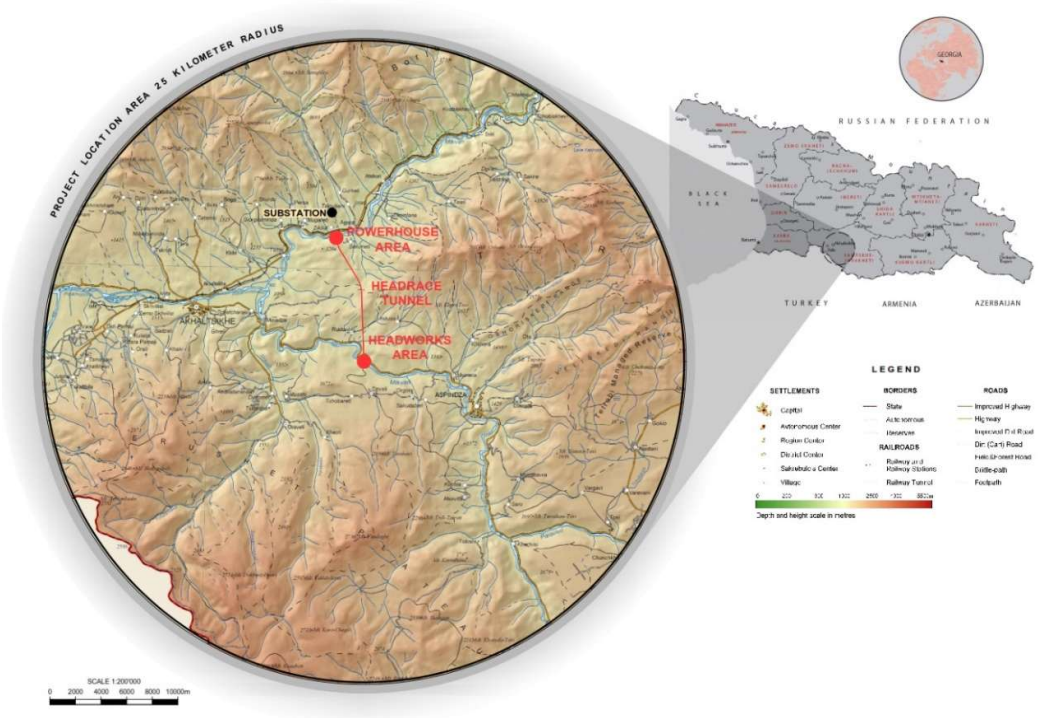


Fig. 1. Mtkvari HPP project location.

1.2 Headworks

The headworks of Mtkvari HPP feature an embankment dam with its crest at 1,019.74 m a.s.l. Maximum flood level, full supply level and minimum operating level for the Mtkvari HPP reservoir are 1,018.04, 1,016.74 and 1,015.74 m a.s.l., respectively. The reservoir allows for a total storage of 6.13 million m^3, with a live storage capacity of 0.53 million m^3 and a flood storage capacity of 0.93 million m^3. The dam is divided into four sections creating two reservoir areas.

- Dam S-1: A 25 m high and 219 m long embankment dam blocks the first bend sleeve on the river. It is constructed as a permeable embankment made of gravel and pebbles fill and acts as a connection between Dam S-2 and the existing road. It divides the reservoir into two areas, which are connected by a culvert.
- Dam S-2: A 28 m high and 374 m long embankment blocks the second bend sleeve on the river and constitutes the main water retaining structure.
- Dam S-3: A 5 m high and 83 m long embankment connects Dam S-2 with the existing appurtenant structures. It is founded on rock and includes a retaining wall acting as a watertight element.
- Dam S-4: A 9 m high lateral embankment with clay fill develops over the natural ridge and is connected to the fish pass by a retaining wall.

The existing appurtenant structures include an ungated surface spillway and a bottom outlet located between Dam S-3 and S-4. The bottom outlet is 127.1 m long, featuring two openings of 6.0 x 6.0 m in size each, equipped with two gates which can be operated only by a gantry crane. The two bottom outlet gates have been upgraded by installing a self-lifting mechanism for each gate to provide more flexibility during impounding and operation.
The ungated surface spillway is located above the bottom outlet and features a box-shaped layout. Its intakes are located on three sides of the structure at an elevation of 1,016.74 m a.s.l., with a total crest length of 46 m.

Following an assessment completed by Gruner Stucky, it was deemed necessary to include a new main service spillway within Dam S-4, located on the left-hand side of the existing outlet works, to optimise the safe discharge of the Design Flood and to handle the Safety Check Flood. Taking this opportunity, a small hydropower plant (SPH) has been also included within the structure of the new main spillway to operate the environmental flow discharged from the dam to produce additional electricity.

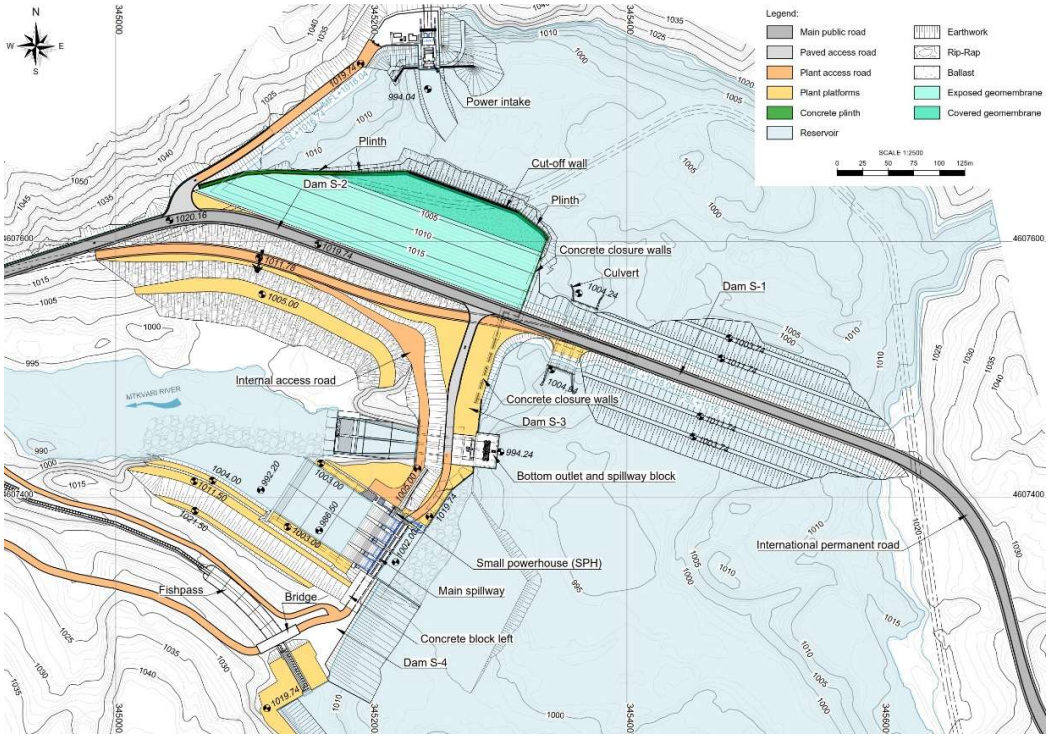


Fig. 2. Mtkvari HPP headworks general arrangement.

1.3 Waterways

The waterways of the scheme feature an intake structure, a 9.6 km long TBM driven headrace tunnel, and a 42.5 m deep surge tank with 19.5 m diameter, which is connected to the top of the headrace tunnel by a 4.5 m diameter vertical shaft. The headrace tunnel bypasses an approximately 27 km long reach of the Mtkvari river. The headrace tunnel was originally lined with 5.26 m diameter precast concrete elements. However, following its construction, remedial lining was carried out in nine sections with a total length of 1,536 m by placing 0.25 m thick additional reinforced concrete. The last 270 m of the headrace tunnel is steel-lined, and it is followed by a surface bifurcation structure and two penstocks embedded in concrete. Each penstock is equipped with an emergency valve.

1.4 Powerhouse

The powerhouse is equipped with two Francis turbines with a rated capacity of 27.04 MW each, while the switchyard is located on the immediate downstream of the powerhouse. The gross head of the scheme can vary from 103.30 m to 105.22 m, and the rated discharge is 32.92 m^3/s per unit.



Fig. 3. Mtkvari HPP headworks (left) and powerhouse (right) construction sites (July 2024).

2 Project design optimisation and dam's safety upgrade

2.1 Dam S-2

Dam Section 2 (S-2) was built up to an elevation of about 1,005 m a.s.l. as an alluvial-fill embankment dam with a central clay core and with a cut-off wall below the core reaching the bedrock. From 1,005 m a.s.l., the design of Dam S-2 was optimised by Gruner Stucky as a geomembrane-face embankment dam by considering these aspects:

- Installation of a geomembrane on the upstream face of Dam S-2 as a sealing element. Above 1,004 m a.s.l., the geomembrane is exposed, while below 1,004 m a.s.l. it is covered by ballast fill. The geomembrane is anchored to the dam crest at 1,019.04 m a.s.l.
- Construction of a reinforced concrete plinth at the upstream toe of the dam embankment, as well as on the left and right banks, where the geomembrane is connected to.
- Implementation of a new cut-off wall in the central riverbed foundation, at the upstream dam toe.
- A closing reinforced concrete wall 0.5 m above and along the left side of Dam S-2. It is founded on sound rock to ensure watertightness at the interface between Dam S-2, Dam S-1 and Dam S-3. The geomembrane is designed to be connected to the crest of this wall.
- A horizontal drainage layer crossing the dam body from upstream to downstream above the clay core to provide a preferred drainage path within the embankment.
- Instrumentation such as leakage detection optical fibres behind the geomembrane, pore pressure cells, standpipe piezometers within the dam body and foundation, as well as a geodetic network, are installed to monitor the behaviour of the dam for the impounding and during its operational life.

Static and dynamic global stability as well as stress-strain analysis were conducted to evaluate the behaviour of the final design solution for Dam S-2 under different load combinations. This was carried out by considering recognised industry guidance (USACE, USBR, etc.).



Fig. 4. Dam S-2 clay core (left) and preparation for the geomembrane of the upstream slope (right).

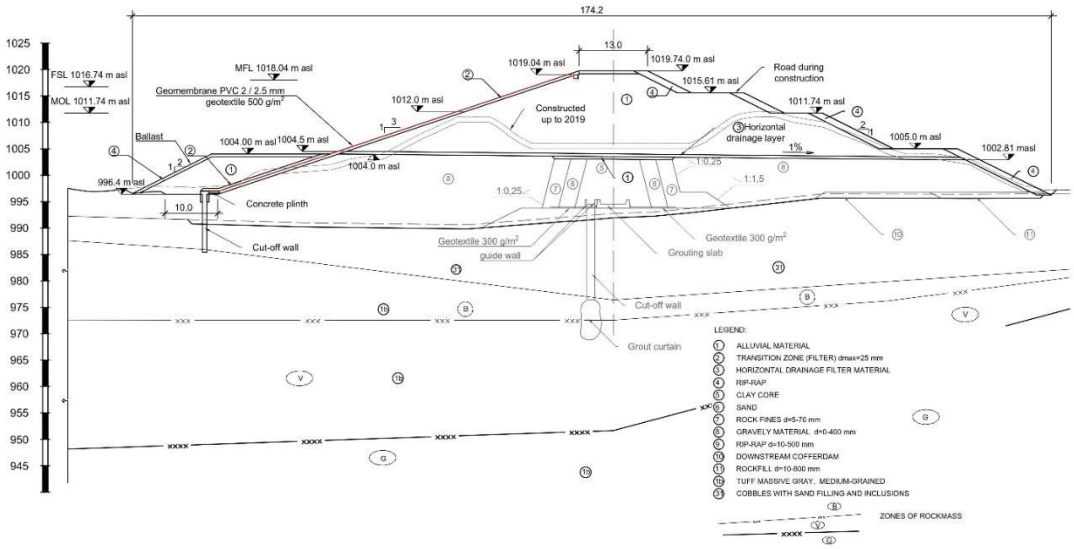


Fig. 5. Final design of typical section of Dam S-2.

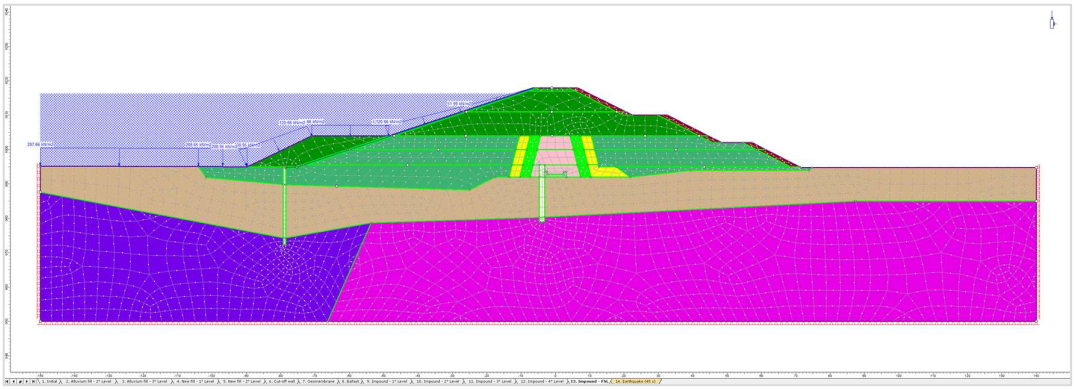


Fig. 6. 2D FEM model of the Dam S-2 for static and dynamic analysis.

2.2 Main spillway

The new main spillway was designed according to ICOLD Bulletin 170, considering a Design Flood with a return period of 10,000 years (1,900 m^3/s), assumed to be released with the gate of highest capacity not being available (n-1), and a Safety Check Flood equal to the Probable Maximum Flood (PMF) (3,400 m^3/s). With the selection of these design criteria, and following an optioneering study, the main spillway location, type, and size were defined by Gruner Stucky (USACE, USBR, etc.).
The main spillway is built within Dam S-4 on the left-hand side of the existing outlet works. It consists of a gated overflow ogee crest control structure with an upstream approach channel, and a downstream chute followed by a stilling basin and exit channel. Main features of its design layout are as follows:

- **Approach channel:** Flow is conveyed in a rectangular channel with riprapped-lined invert.
- **Control structure:** It consists of an overflow ogee crest divided in four (4) 9.0 m wide bays, each equipped with a radial gate. The elevation of the ogee crest is 1,007.25 m a.s.l. A grouting and drainage gallery is located within the ogee structure.
- **Terminal structure:** A reinforced concrete rectangular stilling basin with horizontal floor and sloping end sill is located at the end of the spillway chute to dissipate the energy of the flow and to protect the downstream area from major erosion and scouring.

- **Exit channel:** An excavated exit channel provides a transition between the stilling basin and the main channel of the downstream Mtkvari river. Riprap protection is extended downstream from the channel.

The design posed a significant challenge due to its integration in an existing layout. This led to considerable space restrictions and demanded compliance with the restrictions set by the existing structures, including the existing bottom outlet and fish pass. 3D computational fluid dynamics (CFD) was extensively used to validate the design of the spillway, considering different operational configurations, using the FLOW-3D software package. Particular attention was given to the analysis of the approach flow conditions, validation of the spillway rating curve, and flow behaviour over the ogee crest and chute. Additionally, the analysis included the stilling basin and downstream river flow behaviour.

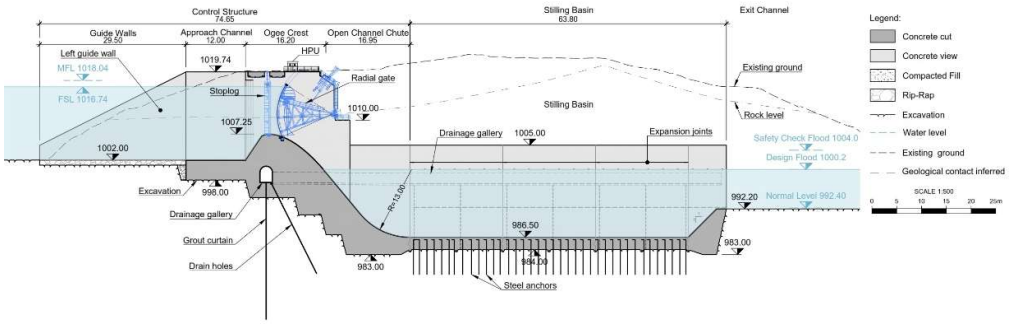


Fig. 7. Longitudinal profile of the new main spillway.



Fig. 8. New main spillway and small powerhouse under construction.

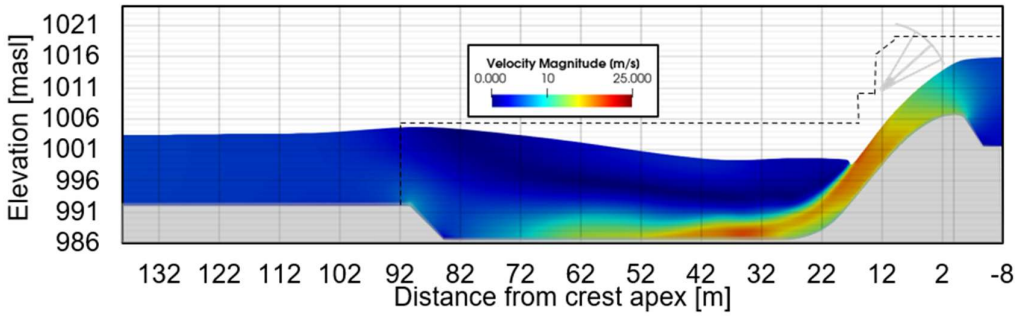


Fig. 9. Flow velocity distribution along middle right spillway bay for Safety Check Flood.

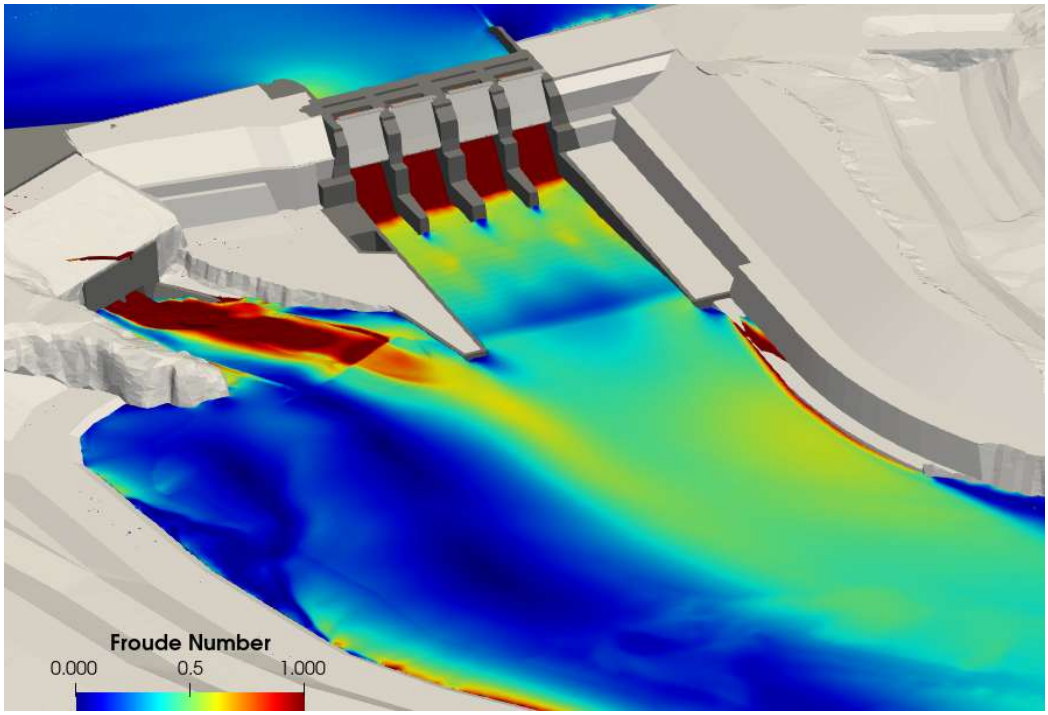


Fig. 10. Distribution of Froude numbers in the spillway and downstream river area for Safety Check Flood.

2.3 Small powerhouse (SPH)

To allow to produce some additional electricity, a small hydropower plant is installed at the spillway. It operates the 5.8 m^3/s constant outflow required to be released from the headworks of Mtkvari HPP as environmental flow.

The operated flow is then released into the stilling basin of the main spillway. Based on the available flow, head and site configuration, a single 1.1 MW vertical Saxo turbine is being installed. The powerhouse is designed as a shaft-type structure integrated within the right-hand sidewall of the control structure of the main spillway.

3 Conclusion

As a result of an open and honest collaboration between the Client, Mtkvari Hesi LLC, the Owner's Engineer, Gruner Stucky, and the Contractor the original design of the project was optimised by means of cost-effective and state-of-the-art technical solutions. This will ensure the sustainable production of renewable energy by means of a safe and reliable power asset, contributing to satisfy the increasing energy demand of the region.

The Author:

Mustafa Aykut is a Civil Engineer and Project Manager specialized in hydropower and dams. He graduated in Civil Engineering from Istanbul Technical University, Turkey. He has been working with Gruner Stucky in design, site supervision and project management activities since 2004 in several countries. He has been the Project Manager for the Mtkvari HPP project, where Gruner Stucky has been acting as the Owner's Engineer.

Sébastien Micheloud is a Civil Engineer and Project Manager specialised in hydropower and dams. He graduated in Civil Engineering (MSc) from École Polytechnique Fédérale de Lausanne, Switzerland. He was involved with Gruner Stucky in the design and site supervision of Vieux Émosson arch dam, in Switzerland. He is currently resident engineer in the Gruner Stucky Caucasus office in Tbilisi, Georgia, where he has been working as Project Manager for some of the major hydropower developments in the country, as well as performing due diligence for hydropower schemes under design and construction.

Giacomo Negrinelli is a Civil Engineer specialised in hydropower, dam safety, river and flood engineering. He graduated in Civil Engineering (BSc) and Environmental Engineering (MSc) from the University of Brescia, Italy. He has been working in research, design and construction roles based in the Netherlands (Deltares), Italy (Frosio Next), UK (Mott MacDonald), Switzerland and Georgia (Gruner Stucky).

Dr Martin Peter Bieri has more than 17 years of experience in hydrological and hydraulic engineering and project management. He obtained his MSc in Civil Engineering from Ecole Polytechnique Fédérale de Lausanne (EPFL) in 2007, his PhD in Hydraulic Engineering from EPFL in 2012 and an MBA from the Asian Institute of Technology (AIT), Bangkok, Thailand in 2019. He worked as a research associate at the Laboratory of Hydraulic Constructions (LCH) at EPFL, Switzerland. In 2017, he joined an international consulting firm based in Switzerland and Thailand. In 2021, he was nominated the head of Gruner Stucky's Major Projects Department. Martin is member of the Project Committee for Mtkvari HPP.

Leontina Galdava is Chief Executive Officer at Mtkvari Hesi LLC. She graduated in Hydropower Engineering (BSc) from the Peoples' Friendship University of Russia, and she obtained a MSc and Doctoral degree from the Georgian Technical University. She has been working in management positions in the Georgian energy sector, as a technical advisor and member of supervisory board for JSC Georgian Energy Development Fund, JSC Partnership Fund, and JSC Georgian State Electrosystem.

Teimuraz Kopadze is Project Manager at Mtkvari Hesi LLC. He graduated from the Tbilisi State University, Georgia, and he obtained a Master's degree in Operations Management from the Free University of Georgia. He is certified as a Project Management Professional by the PMI. He has been working in management positions at JSC Georgian Railways, JSC Partnership Fund, and JSC Nenskra Hydro.

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